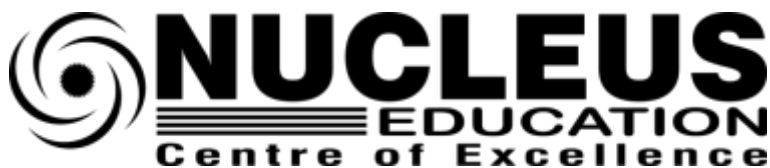


ENTHUSE

IIT MATHEMATICS

**INDEFINITE & DEFINITE INTEGRATION
(SPECIAL DPP)**



RESULTS OF BEST MENTORSHIP BY THE NUCLEUS TEAM



AIR-1
CHITRAANG MURDIA
Gen. Category - 2014



AIR-3
GOVIND LAHOTI
Gen. Category - 2014

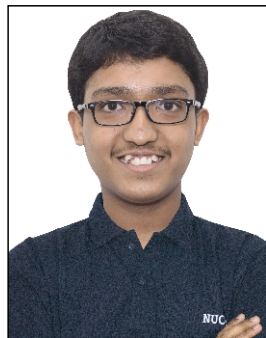


AIR-6
NISHIT AGARWAL
Gen. Category - 2012

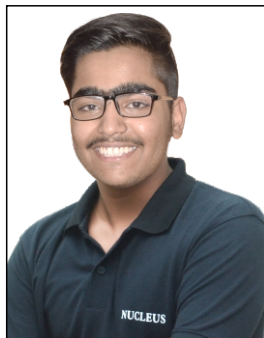


AIR-8
AMEY GUPTA
Gen. Category - 2014

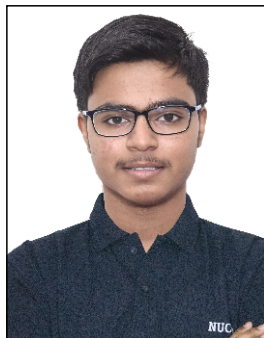
JEE MAIN RESULTS 2020 (January Attempt) OF NUCLEUS EDUCATION



100 Percentile
(Physics)
DAKSH Khandelwal
2020



100 Percentile
(Maths & Physics)
VAIBHAV SAHA
2020



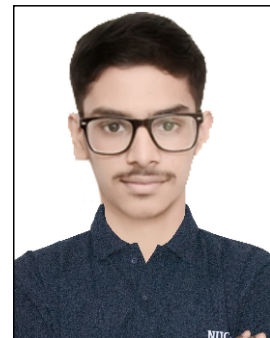
100 Percentile
(Physics)
ANISH MOHAN
2020



100 Percentile
(Maths)
ARCHIT PATNAIK
2020

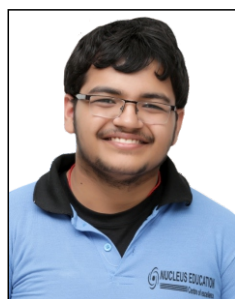


100 Percentile
(Maths)
SWAPNIL YASASVI
2020

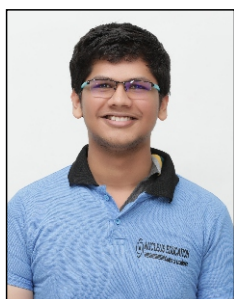


100 Percentile
(Maths)
PARSHANT ARORA
2020

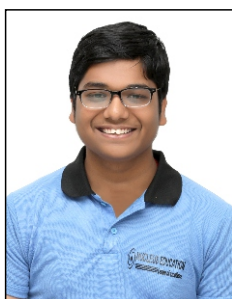
JEE ADVANCED RESULTS OF NUCLEUS EDUCATION



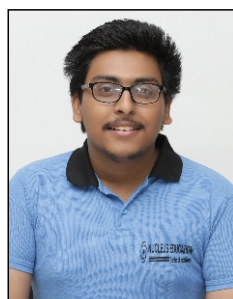
AIR-10
LAKSHAY SHARMA
Gen. Category - 2017



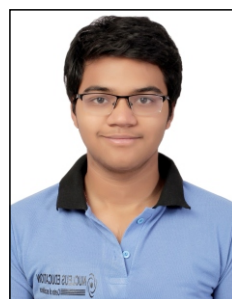
AIR-12
YATEESH AGRAWAL
Gen. Category - 2017



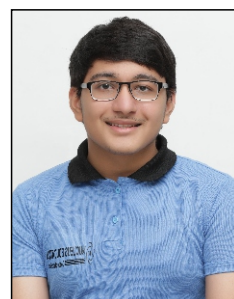
AIR-23
ABHEY GOYAL
Gen. Category - 2017



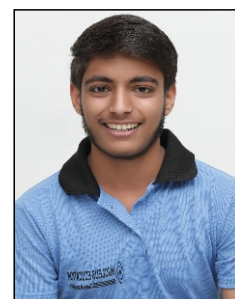
AIR-24
TUSHAR GAUTAM
Gen. Category - 2017



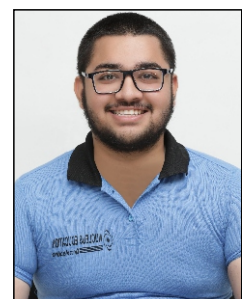
AIR-37
PIYUSH TIBAREWAL
Gen. Category - 2017



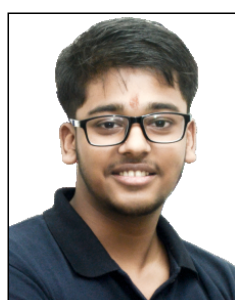
AIR-42
SATVIK
Gen. Category - 2017



AIR-66
MAYANK DUBEY
Gen. Category - 2017



AIR-98
HRITHIK
Gen. Category - 2017



AIR-20
SHASHANK AGRAWAL
Gen. Category - 2018



AIR-27
RAAGHAV RAAJ
Gen. Category - 2018



AIR-32
SHREYA PATHAK
Gen. Category - 2018



AIR-61
SIDDHANT CHOUDAHRY
Gen. Category - 2018



AIR-67
ANISWAR S KRISHNAN
Gen. Category - 2018
DLP



AIR-78
AAYUSH KADAM
Gen. Category - 2018



AIR-61
SARTHAK BEHERA
Gen. Category - 2018
*SDCCP



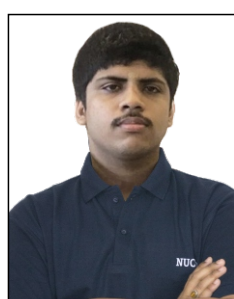
AIR-91
ANDREWS G. VARGHESE
Gen. Category - 2018
*SDCCP



AIR-2
HIMANSHU GAURAV SINGH
Gen. Category - 2019
*SDCCP



AIR-19
VIBHAV AGGARWAL
Gen. Category - 2019



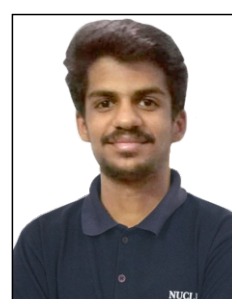
AIR-33
S. PRAJEETH
Gen. Category - 2019
*SDCCP



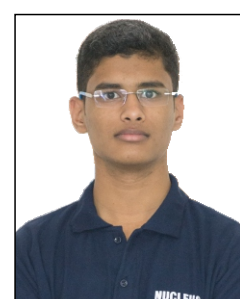
AIR-48
SOHAM MISTRI
Gen. Category - 2019
DLP



AIR-51
SAYANTAN DHAR
Gen. Category - 2019
DLP



AIR-53
GAURAV KRISHAN GUPTA
Gen. Category - 2019
DLP



AIR-86
SATVIK JAIN
Gen. Category - 2019

INDEX

S.No.	Contents	Page No.
INDEFINITE INTEGRATION		
1.	DPP-1	01
2.	DPP-2	02-03
3.	DPP-3	04
4.	DPP-4	05
5.	DPP-5	06
6.	Answer key	07-12
DEFINITE INTEGRATION		
1.	DPP & Answer key	13-14

ELEMENTARY DPP

Find the antiderivative/primitive/integrals of the following by simple manipulation/simplifying and converting them into standard integrals.

1. $\int 2^x \cdot e^x dx$
2. $\int \frac{1 + \cos^2 x}{1 + \cos 2x} dx$
3. $\int \frac{1 - \tan^2 x}{1 + \tan^2 x} dx$
4. $\int \frac{1 + \tan^2 x}{1 + \cot^2 x} dx$
5. $\int \frac{e^{5\ell nx} - e^{4\ell nx}}{e^{3\ell nx} - e^{2\ell nx}} dx$
6. $\int (e^{a\ell nx} + e^{x\ell na}) dx \quad (a > 0)$
7. $\int \frac{\cos 2x}{\cos^2 x \sin^2 x} dx$
8. $\int \frac{1 + 2x^2}{x^2(1 + x^2)} dx$
9. $\int 4 \cos \frac{x}{2} \cdot \cos x \cdot \sin \frac{21}{2} x dx$
10. $\int \frac{\cos x - \sin x}{\cos x + \sin x} (2 + 2 \sin 2x) dx$
11. $\int (3 \sin x \cos^2 x - \sin^3 x) dx$
12. $\int \cos x^\circ dx$
13. $\int \frac{(1+x)^2}{x(1+x^2)} dx$
14. $\int \frac{x}{2x+1} dx$
15. $\int \frac{\sec 2x - 1}{\sec 2x + 1} dx$
16. $\int \frac{2x-1}{x-2} dx$
17. $\int \frac{e^{2x} - 1}{e^x} dx$
18. $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx \quad (\cos x + \sin x > 0)$
19. $\int \frac{\cos 2x - \cos 2\alpha}{\cos x - \cos \alpha} dx$
20. $\int \frac{x^6 - 1}{x^2 + 1} dx$
21. $\int \frac{\sin^3 x + \cos^3 x}{\sin^2 x \cos^2 x} dx$
22. $\int \frac{x^4 + x^2 + 1}{2(1 + x^2)} dx$
23. $\int \frac{\sin^6 x + \cos^6 x}{\sin^2 x \cdot \cos^2 x} dx$
24. $\int \left[\sin^2 \left(\frac{9\pi}{8} + \frac{x}{4} \right) - \sin^2 \left(\frac{7\pi}{8} + \frac{x}{4} \right) \right] dx$
25. $\int \frac{\cos 4x + 1}{\cot x - \tan x} dx$
26. $\int \frac{\sin 2x + \sin 5x - \sin 3x}{\cos x + 1 - 2 \sin^2 2x} dx$
27. $\int \left[\frac{\cot^2 2x - 1}{2 \cot 2x} - \cos 8x \cot 4x \right] dx$
28. $\int \frac{\cos^4 x - \sin^4 x}{\sqrt{1 + \cos 4x}} dx \quad (\cos 2x > 0)$
29. $\int \frac{2x^3 + 3x^2 + 4x + 5}{2x + 1} dx$
30. $\int \frac{(x^2 + \sin^2 x) \sec^2 x}{1 + x^2} dx$
31. $\int \frac{dx}{\sqrt{9 - 16x^2}}$
32. $\int \frac{dx}{25 + 4x^2}$
33. $\int \frac{2x + 3}{3x + 2} dx$
34. $\int \frac{dx}{1 + \sin x}$
35. $\int \frac{\cos 8x - \cos 7x}{1 + 2 \cos 5x} dx$
36. $\int \frac{2 + 3x^2}{x^2(1 + x^2)} dx$
37. $\int \frac{\sin 2x - \sin 2k}{\sin x - \sin k + \cos x - \cos k} dx$
38. $\int \sin x \cos x \cos 2x \cos 4x dx$
39. $\int x^x \ell n(ex) dx$

DPP # 02

(SPECIAL DPP ON INDEFINITE INTEGRATION)

1. $\int \sin x d(\sin x)$
2. $\int \tan^3 x d(\tan x)$
3. $\int \frac{d(1+x^2)}{\sqrt{1+x^2}}$
4. $\int (x+1)^{15} dx$
5. $\int \frac{dx}{(2x-3)^5}$
6. $\int \frac{dz}{(a+bz)^c} (c \neq 1)$
7. $\int \sqrt[5]{(8-3x)^6} dx$
8. $\int \sqrt{8-2x} dx$
9. $\int \frac{m}{\sqrt[3]{(a+bx)^2}} dx$
10. $\int 2x\sqrt{x^2+1} dx$
11. $\int x\sqrt{1-x^2} dx$
12. $\int x^2 \sqrt[5]{x^3+2} dx$
13. $\int \frac{xdx}{\sqrt{x^2+1}}$
14. $\int \frac{x^4 dx}{\sqrt{4+x^5}}$
15. $\int \frac{x^3 dx}{\sqrt[3]{x^4+1}}$
16. $\int \frac{(6x-5)dx}{2\sqrt{3x^2-5x+6}}$
17. $\int \sin^3 x \cos x dx$
18. $\int \frac{\sin x dx}{\cos^2 x}$
19. $\int \frac{\cos x dx}{\sqrt[3]{\sin^2 x}}$
20. $\int \cos^3 x \sin 2x dx$
21. $\int \frac{\sqrt{\ln x}}{x} dx$
22. $\int \frac{(\arctan x)^2 dx}{1+x^2}$
23. $\int \frac{dx}{(\arcsin x)^3 \sqrt{1-x^2}}$
24. $\int \frac{dx}{\cos^2 x \sqrt{1+\tan x}}$
25. $\int \frac{d(1+\ln x)}{\cos^2(1+\ln x)}$
26. $\int (\cos \alpha - \cos 2x) dx$
27. $\int \left[\cos \left(2x - \frac{\pi}{4} \right) \right]^{-2} dx$
28. $\int e^x (\sin e^x) dx$
29. $\int \frac{d(1+x^2)}{1+x^2}$
30. $\int \frac{d(\arcsin x)}{\arcsin x}$
31. $\int \frac{(2x-3)dx}{x^2-3x+8}$
32. $\int \frac{xdx}{x^2+1}$
33. $\int \frac{x^2 dx}{x^3+1}$
34. $\int \frac{e^x dx}{e^x+1}$
35. $\int \frac{e^{2x} dx}{e^{2x}+a^2}$
36. $\int \tan 3x dx$
37. $\int \cot(2x+1) dx$
38. $\int \frac{\sin 2x}{1+\cos^2 x} dx$
39. $\int \frac{dx}{x \ln x}$
40. $\int \frac{(\ln x)^m}{x} dx$
41. $\int e^{\sin x} d(\sin x)$
42. $\int e^{\sin x} \cos x dx$
43. $\int e^{-3x+1} dx$
44. $\int e^{x^2} x dx$
45. $\int e^{-x^3} x^2 dx$
46. $\int \frac{d\left(\frac{x}{3}\right)}{\sqrt{1-\left(\frac{x}{3}\right)^2}}$
47. $\int \frac{dx}{\sqrt{1-25x^2}}$
48. $\int \frac{dx}{1+9x^2}$
49. $\int \frac{dx}{\sqrt{4-x^2}}$
50. $\int \frac{dx}{2x^2+9}$
51. $\int \frac{dx}{\sqrt{4-9x^2}}$

52. $\int \frac{x dx}{x^4 + 1}$

53. $\int \frac{x dx}{\sqrt{a^2 - x^4}}$

54. $\int \frac{x^2 dx}{x^6 + 4}$

55. $\int \frac{x^3 dx}{\sqrt{1 - x^8}}$

56. $\int \frac{e^x dx}{e^{2x} + 4}$

57. $\int \frac{2^x dx}{\sqrt{1 - 4^x}}$

58. $\int \frac{\cos \alpha d\alpha}{a^2 + \sin^2 \alpha}$

59. $\int \frac{e^{2x} - 1}{e^x} dx$

60. $\int (e^x + 1)^3 dx$

61. $\int \frac{1+x}{\sqrt{1-x^2}} dx$

62. $\int \frac{3x-1}{x^2+9} dx$

63. $\int \sqrt{\frac{1-x}{1+x}} dx$

64. $\int \frac{x(1-x^2)}{1+x^4} dx$

65. $\int \frac{1+x-x^2}{\sqrt{(1-x^2)^3}} dx$

66. $\int \frac{dx}{(x + \sqrt{x^2 - 1})^2}$

67. $\int \frac{2x - \sqrt{\arcsin x}}{\sqrt{1-x^2}} dx$

68. $\int \frac{x + (\arccos 3x)^2}{\sqrt{1-9x^2}} dx$

69. $\int \frac{(1+x)^2}{x^2+1} dx$

70. $\int \frac{x^4}{1-x} dx$



DPP # 03

(SPECIAL DPP ON INDEFINITE INTEGRATION)

1. $\int \frac{dx}{x^2 - 7x + 10}$
2. $\int \frac{dx}{x^2 + 3x - 10}$
3. $\int \frac{dx}{4x^2 - 9}$
4. $\int \frac{dx}{2 - 3x^2}$
5. $\int \frac{dx}{(x-1)^2 + 4}$
6. $\int \frac{dx}{x^2 + 2x + 3}$
7. $\int \frac{dx}{x - x^2 - 2.5}$
8. $\int \frac{dx}{4x^2 + 4x + 5}$
9. $\int \frac{dx}{\sqrt{1 - (2x + 3)^2}}$
10. $\int \frac{dx}{\sqrt{4x - 3 - x^2}}$
11. $\int \frac{dx}{\sqrt{8 + 6x - 9x^2}}$
12. $\int \frac{dx}{\sqrt{2 - 6x - 9x^2}}$
13. $\int \cos^2 x dx$
14. $\int \sin^2 x dx$
15. $\int \frac{dx}{1 - \cos x}$
16. $\int \frac{dx}{1 + \sin x}$
17. $\int \frac{1 - \cos x}{1 + \cos x} dx$
18. $\int \frac{1 + \sin x}{1 - \sin x} dx$
19. $\int (\tan^2 x + \tan^4 x) dx$
20. $\int \frac{\cos 2x dx}{1 + \sin x \cos x}$
21. $\int \cos x \sin 3x dx$
22. $\int \cos x \cos 2x \cos 3x dx$
23. $\int \frac{dx}{\cos x}$
24. $\int \frac{1 - \sin x}{\cos x} dx$
25. $\int \frac{\sin^3 x}{\cos x} dx$
26. $\int \frac{\cos^3 x dx}{\sin^4 x}$
27. $\int \frac{\sin^3 \alpha}{\sqrt{\cos \alpha}} d\alpha$
28. $\int \frac{dx}{\cos^4 x}$
29. $\int \tan^4 x dx$
30. $\int \sin^5 x dx$
31. $\int \sin^4 x dx$
32. $\int \tan^3 x dx$
33. $\int \frac{dx}{\sin^6 x}$
34. $\int (x+1)\sqrt{x^2 + 2x} dx$
35. $\int \frac{e^{\sqrt{x}}}{\sqrt{x}} dx$
36. $\int \sqrt{1 - e^x} e^x dx$
37. $\int \frac{2x^5 - 3x^2}{1 + 3x^3 - x^6} dx$
38. $\int \frac{\sqrt{x} dx}{1 + x^{\frac{3}{2}}}$
39. $\int \frac{2x + 3}{\sqrt{1 + x^2}} dx$
40. $\int \frac{dx}{x\sqrt{3 - \ln^2 x}}$
41. $\int \frac{x^2 - x + 1}{\sqrt{(x^2 + 1)^3}} dx$
42. $\int \frac{(\arctan x)^n}{1 + x^2} dx$
43. $\int \frac{d\phi}{\sin^2 \phi \cos^2 \phi}$
44. $\int \frac{x^3 dx}{x + 1}$
45. $\int \frac{xdx}{(x+1)^3}$
46. $\int \frac{(x+2)dx}{x^2 + 2x + 2}$
47. $\int \frac{(x-3)dx}{\sqrt{3 - 2x - x^2}}$
48. $\int \frac{(3x-1)dx}{\sqrt{x^2 + 2x + 2}}$
49. $\int \frac{\cot x}{\ln(\sin x)} dx$
50. $\int \frac{e^x - 1}{e^x + 1} dx$
51. $\int e^{e^x + x} dx$
52. $\int e^{2x^2 + \ln x} dx$
53. $\int \frac{\sin 2x dx}{4 - \cos^2 2x}$
54. $\int \frac{d\phi}{\sqrt{3} \cos \phi + \sin \phi}$

DPP # 04

(SPECIAL DPP ON INDEFINITE INTEGRATION)

PART-I

1. $\int x e^{-x} dx$
2. $\int x^n \ln x dx \quad (n \neq -1)$
3. $\int \arctan \sqrt{x} dx$
4. $\int \frac{x \arctan x}{\sqrt{1+x^2}} dx$
5. $\int \frac{\arcsin \sqrt{x}}{\sqrt{1-x}} dx$
6. $\int \frac{x^2 dx}{(1+x^2)^2}$
7. $\int \frac{x^3 dx}{\sqrt{1+x^2}}$
8. $\int \frac{\ln^3 x}{x^2} dx$
9. $\int (\arctan x)^2 x dx$
10. $\int e^{ax} \cos nx dx$
11. $\int \sin(\ln x) dx$
12. $\int \frac{x^2 dx}{\sqrt{1-x^2}}$
13. $\int \frac{3x^2-1}{2x\sqrt{x}} \arctan x dx$
14. $\int \ln(x + \sqrt{1+x^2}) dx$

PART-II

15. $\int \frac{dx}{1+\sqrt{x+1}}$
16. $\int \frac{4x+3}{(x-2)^3} dx$
17. $\int \frac{dx}{1+\sqrt[3]{x+1}}$
18. $\int \frac{\sqrt{x} dx}{\sqrt{x} - \sqrt[3]{x}}$
19. $\int \frac{\sqrt{1+x^2}}{x^4} dx$
20. $\int \frac{dx}{\sqrt{(a^2+x^2)^3}}$
21. $\int \frac{dx}{x^2 \sqrt{x^2-9}}$
22. $\int \frac{dx}{x \sqrt{1+x^2}}$
23. $\int \frac{dx}{\sqrt{x-x^2}}$
24. $\int \frac{(x+1)dx}{x(1+xe^x)}$
25. $\int \frac{x^4 dx}{\sqrt{(1-x^2)^3}}$
26. $\int \frac{dx}{x^4 \sqrt{x^2+4}}$
27. $\int \frac{dx}{x^4 \sqrt{x^2-3}}$

DPP # 05

(SPECIAL RACE ON INDEFINITE INTEGRATION)

1. $\int \frac{dx}{6x^3 - 7x^2 - 3x}$

2. $\int \frac{x dx}{x^4 - 3x^2 + 2}$

3. $\int \frac{(2x^2 - 5)dx}{x^4 - 5x^2 + 6}$

4. $\int \frac{dx}{x^4 - x^2}$

5. $\int \frac{3x^2 + 1}{(x^2 - 1)^3} dx$

6. $\int \frac{xdx}{x^3 - 1}$

7. $\int \frac{x^2 dx}{1 - x^4}$

8. $\int \frac{dx}{(1 + x^2)^4}$

9. $\int \frac{x^3 - 1}{4x^3 - x} dx$

10. $\int \left(\frac{x+2}{x-1} \right)^2 \frac{dx}{x}$

11. $\int \frac{x^2 dx}{(x+2)^2 (x+4)^2}$

12. $\int \frac{dx}{\sqrt[4]{(x-1)^3 (x+2)^5}}$

13. $\int \frac{dx}{1 - \sin^4 x}$

14. $\int \sqrt[3]{\frac{\sin^2 x}{\cos^{14} x}} dx$

15. $\int \frac{1 + x^4}{(1 - x^4)^{3/2}} dx$

16. $\int \frac{dx}{\cos x \cdot \sin^3 x}$

17. $\int \frac{dx}{\cos^3 x \cdot \sin^3 x}$

18. $\int \frac{\sin x dx}{(1 - \cos x)^2}$

19. $\int \frac{\cos x dx}{(1 - \cos x)^2}$

20. $\int \frac{dx}{\sqrt[4]{\sin^3 x \cos^5 x}}$

21. $\int \frac{x^4 dx}{x^{15} - 1}$

22. $\int \frac{\sin 2x dx}{\cos^4 x + \sin^4 x}$

23. $\int \frac{(x^2 - 1)dx}{x\sqrt{x^4 + 3x^2 + 1}}$

24. $\int \frac{(e^{3x} + e^x)dx}{e^{4x} - e^{2x} + 1}$

25. $\int \frac{(x + \sin x)dx}{1 + \cos x}$

26. $\int \frac{x^2 - 1}{x^2 + 1} \cdot \frac{dx}{\sqrt{1 + x^4}}$

27. $\int e^{\sin x} \frac{x \cos^3 x - \sin x}{\cos^2 x} dx$

28. $\int \frac{x^5 + x^4 - 8}{x^3 - 4x} dx$

29. $\int \frac{(x + \sqrt{1 + x^2})^{15}}{\sqrt{1 + x^2}} \cdot dx$

30. $\int \frac{\sin^3 x \cdot dx}{(\cos^4 x + 3\cos^2 x + 1) \tan^{-1}(\sec x + \cos x)}$

31. $\int \frac{dx}{5 - 4\sin x + 3\cos x}$

32. $\int \frac{dx}{4 - 3\cos^2 x + 5\sin^2 x}$

33. $\int \frac{2x^2 + 41x - 91}{(x-1)(x+3)(x-4)} dx$

34. $\int \frac{dx}{(\sin x + 2\sec x)^2}$

35. $\int \frac{(\cos 2x - 3)dx}{\cos^4 x \sqrt{4 - \cot^2 x}}$

36. $\int \frac{dx}{\sin \frac{x}{2} \sqrt{\cos^3 \frac{x}{2}}}$

37. $\int \sqrt{e^x - 1} dx$

38. $\int \frac{dx}{\cos^3 x \sqrt{\sin 2x}}$

ANSWER KEY

DPP # 01

1. $\frac{2^x \cdot e^x}{1 + \ln 2} + C$ 2. $\frac{1}{2}(\tan x + x) + C$ 3. $\frac{1}{2} \sin 2x + C$ 4. $\tan x - x + C$
5. $\frac{x^3}{3} + C$ 6. $\frac{x^{a+1}}{a+1} + \frac{a^x}{\ln a} + C$ 7. $-(\cot x + \tan x) + C$
8. $-\frac{1}{x} + \tan^{-1} x + C$ 9. $-\left[\frac{1}{9} \cos 9x + \frac{1}{10} \cos 10x + \frac{1}{11} \cos 11x + \frac{1}{12} \cos 12x \right] + C$
10. $\sin 2x + C$ 11. $-\frac{\cos 3x}{3} + C$ 12. $\frac{180}{\pi} \sin x^\circ + C$ 13. $\ln x + 2 \tan^{-1} x + C$
14. $\frac{1}{2} \left[x - \frac{\ln(2x+1)}{2} \right] + C$ 15. $\tan x - x + C$ 16. $2x + 3 \ln(x-2) + C$
17. $e^x + e^{-x} + C$ 18. $x + C$ 19. $2(\sin x + x \cos \alpha) + C$
20. $\frac{x^5}{5} - \frac{x^3}{3} + x - 2 \tan^{-1} x + C$ 21. $\sec x - \operatorname{cosec} x + C$ 22. $\frac{1}{2} \left[\frac{x^3}{3} + \tan^{-1} x \right] + C$
23. $\tan x - \cot x - 3x + C$ 24. $-\sqrt{2} \cos \frac{x}{2} + C$ 25. $-\frac{\cos 4x}{8} + C$
26. $-2 \cos x + C$ 27. $-\frac{\cos 8x}{8} + C$ 28. $\frac{x}{\sqrt{2}} + C$
29. $\frac{x^3}{3} + \frac{x^2}{2} + \frac{3x}{2} + \frac{7}{4} \ln(2x+1)$ 30. $\tan x - \tan^{-1} x + C$ 31. $\frac{1}{4} \sin^{-1} \frac{4}{3} x + C$
32. $\frac{1}{10} \tan^{-1} \frac{2x}{5} + C$ 33. $\frac{2}{3} x + \frac{5}{9} \ln(3x+2) + C$ 34. $\tan x - \sec x + C$
35. $\frac{\sin 3x}{3} - \frac{\sin 2x}{2} + C$ 36. $-\frac{2}{x} + \tan^{-1} x + C$ 37. $(\sin x - \cos x) + (\sin k + \cos k)x + C$
38. $-\frac{1}{64} \cos 8x + C$ 39. $x^x + C$

DPP # 02

1. $\frac{\sin^2 x}{2} + C$
2. $\frac{\tan^4 x}{4} + C$
3. $2\sqrt{1+x^2} + C$
4. $\frac{(x+1)^{16}}{16} + C$
5. $C - \frac{1}{8(2x-3)^4}$
6. $\frac{(a+bz)^{1-c}}{b(1-c)} + C$
7. $C - \frac{5}{33}(8-3x)^{11/5}$
8. $C - \frac{\sqrt{(8-2x)^3}}{3}$
9. $\frac{3m}{b}\sqrt[3]{a+bx} + C$
10. $\frac{2}{3}\sqrt{(x^2+1)^3} + C$
11. $C - \frac{1}{3}\sqrt{(1-x^2)^3}$
12. $\frac{5}{18}\sqrt[5]{(x^3+2)^6} + C$
13. $\sqrt{x^2+1} + C$
14. $\frac{2}{5}\sqrt{.4+x^5} + C$
15. $\frac{3}{8}\sqrt[3]{(x^4+1)^2} + C$
16. $\sqrt{3x^2-5x+6} + C$
17. $\frac{1}{4}\sin^4 x + C$
18. $\sec x + C$
19. $3\sqrt[3]{\sin x} + C$
20. $C - \frac{2}{5}\cos^5 x$
21. $\frac{2}{3}\sqrt{(\ln x)^3} + C$
22. $\frac{(\arctan x)^3}{3} + C$
23. $C - \frac{1}{2(\arcsin x)^2}$
24. $2\sqrt{1+\tan x} + C$
25. $\tan(1 + \ln x) + C$
26. $x \cos \alpha - \frac{1}{2}\sin 2x + C$
27. $\frac{1}{2}\tan\left(2x - \frac{\pi}{4}\right) + C$ or $\frac{1}{2}(\tan 4x - \sec 4x) + C$
28. $C - \cos(e^x)$
29. $\ln(1+x^2) + C$
30. $\ln |\arcsin x| + C$
31. $\ln(x^2 - 3x + 8) + C$
32. $\frac{1}{2}\ln(x^2+1) + C$
33. $\frac{1}{3}\ln |x^3+1| + C$
34. $\ln(e^x+1) + C$
35. $\frac{1}{2}\ln(e^{2x}+a^2) + C$
36. $C - \frac{1}{3}\ln |\cos 3x|$
37. $\frac{1}{2}\ln |\sin(2x+1)| + C$
38. $C - \ln(1+\cos^2 x)$
39. $\ln |\ln x| + C$
40. $\frac{\ln^{m+1} x}{m+1} + C$ if $m \neq -1$ and $\ln |\ln x| + C$ if $m = -1$
41. $e^{\sin x} + C$
42. $e^{\sin x} + C$
43. $C - \frac{e^{1-3x}}{3}$
44. $0.5e^{x^2} + C$
45. $C - \frac{1}{3}e^{-x^3}$
46. $\arcsin \frac{x}{3} + C$
47. $\frac{1}{5}\arcsin 5x + C$
48. $\frac{1}{3}\arctan 3x + C$
49. $\arcsin \frac{x}{2} + C$
50. $\frac{1}{3\sqrt{2}}\arctan \frac{\sqrt{2}}{3}x + C$
51. $\frac{1}{3}\arcsin \frac{3x}{2} + C$
52. $\frac{1}{2}\arctan x^2 + C$
53. $\frac{1}{2}\arcsin \frac{x^2}{a} + C$
54. $\frac{1}{6}\arctan \frac{x^3}{2} + C$
55. $\frac{1}{4}\arcsin x^4 + C$
56. $\frac{1}{2}\arctan \frac{e^x}{2} + C$
57. $\frac{\arcsin 2^x}{\ln 2} + C$
58. $\frac{1}{a}\arctan \frac{\sin \alpha}{a} + C$
59. $e^x + e^{-x} + C$

60. $\frac{1}{3}e^{3x} + \frac{3}{2}e^{2x} + 3e^x + x + C$
61. $\arcsin x - \sqrt{1-x^2} + C$
62. $\frac{3}{2}\ln(x^2+9) - \frac{1}{3}\arctan \frac{x}{3} + C$
63. $\arcsin x + \sqrt{1-x^2} + C$
64. $\frac{1}{2}\arctan x^2 - \frac{1}{4}\ln(x^4+1) + C$
65. $\arcsin x + \frac{1}{\sqrt{1-x^2}} + C$
66. $\frac{2}{3}[x^3 - \sqrt{(x^2-1)^3}] - x + C$
67. $C - 2\sqrt{1-x^2} - \frac{2}{3}\sqrt{(\arcsin x)^3}$
68. $C - \frac{1}{9}[\sqrt{1-9x^2} + (\arccos 3x)^3]$
69. $x + \ln(x^2+1) + C$
70. $C - \frac{1}{4}x^4 - \frac{1}{3}x^3 - \frac{1}{2}x^2 - x - \ln|1-x|$

DPP # 03

1. $\frac{1}{3}\ln\left|\frac{x-5}{x-2}\right| + C$
2. $\frac{1}{7}\ln\left|\frac{x-2}{x+5}\right| + C$
3. $\frac{1}{12}\ln\left|\frac{2x-3}{2x+3}\right| + C$
4. $\frac{1}{2\sqrt{6}}\ln\left|\frac{\sqrt{2}+x\sqrt{3}}{\sqrt{2}-x\sqrt{3}}\right| + C$
5. $\frac{1}{2}\arctan \frac{x-1}{2} + C$
6. $\frac{1}{\sqrt{2}}\arctan \frac{x+1}{\sqrt{2}} + C$
7. $\frac{2}{3}\arctan \frac{1-2x}{3} + C$
8. $\frac{1}{4}\arctan \frac{2x+1}{2} + C$
9. $\frac{1}{2}\arcsin(2x+3) + C$
10. $\arcsin(x-2) + C$
11. $\frac{1}{3}\arcsin \frac{3x-1}{3} + C$
12. $\frac{1}{3}\arcsin \frac{3x+1}{\sqrt{3}} + C$
13. $\frac{x}{2} + \frac{\sin 2x}{4} + C$
14. $\frac{x}{2} - \frac{\sin 2x}{4} + C$
15. $C - \cot \frac{x}{2}$
16. $\tan\left(\frac{x}{2} - \frac{\pi}{4}\right) + C$
17. $2\tan \frac{x}{2} - x + C$
18. $2\tan\left(\frac{x}{2} + \frac{\pi}{4}\right) - x + C$
19. $\frac{1}{3}\tan^3 x + C$
20. $\ln(2 + \sin 2x) + C$
21. $C - \frac{1}{4}\left(\frac{\cos 4x}{2} + \cos 2x\right)$
22. $\frac{1}{8}\left(2x + \sin 2x + \frac{1}{2}\sin 4x + \frac{1}{3}\sin 6x\right) + C$
23. $\ln\left|\tan\left(\frac{\pi}{4} + \frac{x}{2}\right)\right| + C$
24. $\ln(1 + \sin x) + C$
25. $\frac{\cos^2 x}{2} - \ln|\cos x| + C$
26. $\frac{1}{\sin x} - \frac{1}{3\sin^3 x} + C$
27. $2\sqrt{\cos \alpha}\left(\frac{\cos^2 \alpha}{5} - 1\right) + C$
28. $\tan x + \frac{1}{3}\tan^3 x + C$
29. $\frac{1}{3}\tan^3 x - \tan x + x + C$
30. $C - \cos x + \frac{2}{3}\cos^3 x - \frac{1}{5}\cos^5 x$
31. $\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$
32. $\frac{1}{2}\tan^2 x + \ln|\cos x| + C$
33. $C - \cot x - \frac{2}{3}\cot^3 x - \frac{1}{5}\cot^5 x$
34. $\frac{1}{3}\sqrt{(x^2+2x)^3} + c$
35. $2e^{\sqrt{x}} + c$

36. $c - \frac{2}{3}(1 - e^x)^{\frac{3}{2}}$

37. $c - \frac{1}{3} \ln |1 + 3x^3 - x^6|$

38. $\frac{2}{3} \ln(1 + x^{\frac{3}{2}}) + c$

39. $2\sqrt{1+x^2} + 3\ln(x + \sqrt{1+x^2}) + c$

40. $\arcsin \frac{\ln x}{\sqrt{3}} + c$

41. $\frac{1}{\sqrt{x^2+1}} + \ln(x + \sqrt{x^2+1}) + c$

42. $\frac{(\arctan x)^{n+1}}{n+1} + c$ if $n \neq -1$ and $\ln|\arctan x|$ if $n = -1$

43. $c - 2 \cot 2\phi$

44. $\frac{x^3}{3} - \frac{x^2}{2} + x - \ln|x+1| + C$

45. $\frac{-1}{x+1} + \frac{1}{2(x+1)^2} + C$

46. $\frac{1}{2} \ln(x^2 + 2x + 2) + \arctan(x+1) + c$

47. $c - \sqrt{3-2x-x^2} - 4 \arcsin \frac{x+1}{2}$

48. $3\sqrt{x^2+2x+2} - 4\ln(x+1+\sqrt{x^2+2x+2}) + c$

49. $\ln|\ln(\sin x)| + C$

50. $2\ln(e^{x/2} + e^{-x/2}) + C$

51. $e^{e^x} + C$

52. $\frac{1}{4}e^{2x^2} + c$

53. $C - \frac{1}{8} \ln \frac{2+\cos 2x}{2-\cos 2x}$

54. $\frac{1}{2} \ln \left| \tan \left(\frac{\phi}{2} + \frac{\pi}{6} \right) \right| + c$

DPP # 04

PART-I

1. $C - e^{-x}(x+1)$

2. $\frac{x^{n+1}}{n+1} \left(\ln x - \frac{1}{n+1} \right) + C$

3. $x \arctan \sqrt{x} - \sqrt{x} + \arctan \sqrt{x} + C$

4. $\sqrt{1+x^2} \arctan x - \ln(x + \sqrt{1+x^2}) + C$

5. $2(\sqrt{x} - \sqrt{1-x} \arcsin \sqrt{x}) + C$

6. $C - \frac{x}{2(1+x^2)} + \frac{1}{2} \arctan x$

7. $x^2 \sqrt{1+x^2} - \frac{2}{3} \sqrt{(1+x^2)^3} + C$

8. $C - \frac{1}{x} (\ln^3 x + 3\ln^2 x + 6\ln x + 6)$

9. $\frac{x^2+1}{2} (\arctan x)^2 - x \arctan x + \frac{1}{2} \ln(1+x^2) + C$

10. $\frac{e^{ax}}{a^2+n^2} (n \sin nx + a \cos nx) + C$

11. $\frac{x}{2} (\sin \ln x - \cos \ln x) + C$

12. $C - \frac{x}{2} \sqrt{1-x^2} + \frac{1}{2} \arcsin x$

13. $\frac{(x^2+1) \arctan x}{\sqrt{x}} - 2\sqrt{x} + C$

14. $x \ln(x + \sqrt{1+x^2}) - \sqrt{1+x^2} + C$

PART-II

15. $2\left[\sqrt{x+1} - \ln(1+\sqrt{x+1})\right] + C$

16. $C - \frac{11}{2(x-2)^2} - \frac{4}{x-2}$

17. $\frac{3}{2}(x+1)^{2/3} - 3(x+1)^{1/3} + 3\ln|1+\sqrt[3]{x+1}| + C$

18. $x + \frac{6\sqrt[6]{x^5}}{5} + \frac{3\sqrt[3]{x^2}}{2} + 2\sqrt{x} + 3\sqrt[3]{x} + 6\sqrt[6]{x} + 6\ln|\sqrt[6]{x}-1| + C$

19. $c - \frac{\sqrt{(1+x^2)^3}}{3x^3}$

20. $\frac{x}{a^2\sqrt{x^2+a^2}} + c$

21. $\frac{\sqrt{x^2-9}}{9x} + c$

22. $\ln \frac{|x|}{1+\sqrt{x^2+1}} + c$

23. $2 \arcsin \sqrt{x} + C$

24. $\ln \left| \frac{xe^x}{1+xe^x} \right| + C$

25. $C - \frac{x(x^2-3)}{2\sqrt{1-x^2}} - \frac{3}{2} \arcsin x$

26. $\frac{\sqrt{4+x^2}(x^2-2)}{24x^3} + C$

27. $\frac{\sqrt{x^2-3}(2x^2+3)}{27x^3} + c$

DPP # 05

1. $\frac{3}{11} \ln|3x+1| + \frac{2}{33} \ln|2x-3| - \frac{1}{3} \ln|x| + c$

2. $\ln \sqrt{\frac{x^2-2}{x^2-1}} + C$

3. $\frac{1}{2\sqrt{2}} \ln \left| \frac{x-\sqrt{2}}{x+\sqrt{2}} \right| + \frac{1}{2\sqrt{3}} \ln \left| \frac{x-\sqrt{3}}{x+\sqrt{3}} \right| + c$

4. $\frac{1}{x} + \frac{1}{2} \ln \left| \frac{x-1}{x+1} \right| + c$

5. $c - \frac{x}{(x^2-1)^2}$

6. $\frac{1}{3} \ln \frac{|x-1|}{\sqrt{x^2+x+1}} + \frac{1}{\sqrt{3}} \arctan \frac{2x+1}{\sqrt{3}} + c$

7. $\frac{1}{4} \ln \left| \frac{1+x}{1-x} \right| - \frac{1}{2} \arctan x + c$

8. $\frac{15x^5 + 40x^3 + 33x}{48(1+x^2)^3} + \frac{15}{48} \arctan x + C$

9. $\frac{1}{4}x + \ln|x| - \frac{7}{16} \ln|2x-1| - \frac{9}{16} \ln|2x+1| + c$

10. $4\ln|x| - 3\ln|x-1| - \frac{9}{x-1} + C$

11. $2\ln \left| \frac{x+4}{x+2} \right| - \frac{5x+12}{x^2+6x+8} + c$

12. $\frac{4}{3} \sqrt[4]{\frac{x-1}{x+2}} + C$

13. $\frac{1}{2} \tan x + \frac{1}{2\sqrt{2}} \arctan(\sqrt{2} \tan x) + C$

14. $\frac{3}{55} \sqrt[3]{\tan^5 x (5 \tan^2 x + 11)} + c$

15. $\frac{1}{\sqrt{\frac{1}{x^2} - x^2}} + c$

16. $\ln|\tan x| - \frac{1}{2\sin^2 x} + c$

17. $\frac{1}{2}(\tan^2 x - \cot^2 x) + 2\ln|\tan x| + c$

18. $\frac{1}{\cos x - 1} + c$

19. $\frac{1}{2} \cot \frac{x}{2} - \frac{1}{6} \cot^3 \frac{x}{2} + C$

20. $4\sqrt[4]{\tan x} + c$
21. $\frac{1}{15} \left[\frac{1}{2} \ln \frac{(z-1)^2}{z^2+z+1} - \sqrt{3} \arctan \frac{2z+1}{\sqrt{3}} \right] + c$, where $z = x^5$
22. $\arctan(\tan^2 x) + c$
23. $\ln \frac{x^2+1+\sqrt{x^4+3x^2+1}}{x} + C$
24. $\arctan(e^x - e^{-x}) + C$
25. $x \tan \frac{x}{2} + C$
26. $\frac{1}{\sqrt{2}} \arccos \frac{x\sqrt{2}}{x^2+1} + C$
27. $e^{\sin x} (x - \sec x) + c$
28. $\frac{x^3}{3} + \frac{x^2}{2} + 4x + \ln \left| \frac{x^2(x-2)^5}{(x+2)^3} \right| + c$
29. $\frac{(\sqrt{1+x^2} + x)^{15}}{15} + c$
30. $\log(\tan^{-1}(\cos x + \sec x)) + c$
31. $\frac{1}{2 - \tan \frac{x}{2}} + c$
32. $\frac{1}{3} \arctan(3 \tan x) + C$
33. $\ln \left| \frac{(x-1)^4(x-4)^5}{(x+3)^7} \right| + c$
34. $\frac{\cos 2x - 15}{15(4 + \sin 2x)} + \frac{4}{15\sqrt{15}} \arcsin \frac{4 \sin 2x + 1}{4 + \sin 2x} + C$
or
 $\frac{z}{8a^2(z^2+a^2)} + \frac{1}{8a^3} \tan^{-1} \frac{z}{a} + c$, where $z = \tan x + \frac{1}{4}$ & $a^2 = \frac{15}{16}$
35. $C - \frac{1}{3} \tan x (2 + \tan^2 x) \sqrt{4 - \cot^2 x}$
36. $\frac{4}{\sqrt{\cos \frac{x}{2}}} + 2 \arctan \sqrt{\cos \frac{x}{2}} - \ln \frac{1 + \sqrt{\cos \frac{x}{2}}}{1 - \sqrt{\cos \frac{x}{2}}} + C$
37. $2\sqrt{e^x - 1} - 2 \arctan \sqrt{e^x - 1} + C$
38. $\frac{\sqrt{2}}{5} (\tan^2 x + 5) \sqrt{\tan x} + C$

Special DPP On Definite Integration

Evaluate the following definite integrals.

1. $\int_0^1 \frac{\sin^{-1} \sqrt{x}}{\sqrt{x}(1-x)} dx$
2. $\int_0^{\ln 2} x e^{-x} dx$
3. $\int_1^e \left(\frac{1}{\sqrt{x} \ln x} + \sqrt{\frac{\ln x}{x}} \right) dx$
4. Given $f'(x) = \frac{\cos x}{x}$, $f\left(\frac{\pi}{2}\right) = a$, $f\left(\frac{3\pi}{2}\right) = b$. Find the value of the definite integral $\int_{\pi/2}^{3\pi/2} f(x) dx$.
5. $\int_{-1}^1 \frac{x dx}{\sqrt{5-4x}}$
6. $\int_2^e \left(\frac{1}{\ell \ln x} - \frac{1}{\ell n^2 x} \right) dx$
7. $\int_0^{\pi/4} \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx$
8. $\int_0^{\pi/2} \frac{\cos x dx}{(1+\sin x)(2+\sin x)}$
9. $\int_0^{\pi/4} \frac{\sin^2 x \cdot \cos^2 x}{(\sin^3 x + \cos^3 x)^2} dx$
10. $\int_{1/3}^3 \frac{\sin^{-1} \frac{x}{\sqrt{1+x^2}}}{x} dx$
11. $\int_2^3 \frac{dx}{\sqrt{(x-1)(5-x)}}$
12. $\int_{3/2}^2 \left(\frac{x-1}{3-x} \right)^{1/2} dx$
13. $\int_0^{\pi/4} (x \cos x \cdot \cos 3x) dx$
14. $\int_0^{\pi/2} \frac{dx}{5+4 \sin x}$
15. $\int_2^3 \frac{dx}{(x-1)\sqrt{x^2-2x}}$
16. $\int_0^{\pi/2} \frac{dx}{1+\cos \theta \cdot \cos x}$ $\theta \in (0, \pi)$
17. $\int_0^{\ln 3} \frac{e^x + 1}{e^{2x} + 1} dx$
18. $\int_0^{\pi/4} \cos 2x \sqrt{1-\sin 2x} dx$
19. $\int_0^3 \sqrt{\frac{x}{3-x}} dx$
20. $\int_0^{1/2} \frac{dx}{(1-2x^2)\sqrt{1-x^2}}$
21. $\int_1^2 \frac{dx}{x(x^4+1)}$
22. $\int_0^{\pi/2} \sin \phi \cos \phi \sqrt{a^2 \sin^2 \phi + b^2 \cos^2 \phi} d\phi$ $a \neq b$ ($a > 0, b > 0$)
23. (a) $\int_0^{3\pi/4} ((1+x) \sin x + (1-x) \cos x) dx$ (b) $\int_{\pi/2}^{\pi} x^{\sin x} (1+x \cos x \cdot \ln x + \sin x) dx$
24. $\int_0^1 x (\tan^{-1} x)^2 dx$
25. Suppose that f, f' and f'' are continuous on $[0, \ln 2]$ and that $f(0) = 0, f'(0) = 3, f(\ln 2) = 6, f'(\ln 2) = 4$ and $\int_0^{\ln 2} e^{-2x} \cdot f(x) dx = 3$. Find the value of $\int_0^{\ln 2} e^{-2x} \cdot f''(x) dx$.
26. $\int_0^1 \frac{dx}{x^2 + 2x \cos \alpha + 1}$ where $-\pi < \alpha < \pi$
27. $\int_a^b \frac{dx}{\sqrt{1+x^2}}$ where $a = \frac{e-e^{-1}}{2}$ & $b = \frac{e^2-e^{-2}}{2}$
28. $\int_0^1 \frac{1-x^2}{1+x^2+x^4} dx$
29. $\int_0^1 x^5 \sqrt{\frac{1+x^2}{1-x^2}} dx$
30. Suppose that the function f, g, f' and g' are continuous over $[0, 1]$, $g(x) \neq 0$ for $x \in [0, 1]$, $f(0) = 0$, $g(0) = \pi$, $f(1) = \frac{2009}{2}$ and $g(1) = 1$. Find the value of the definite integral, $\int_0^1 \frac{f(x) \cdot g'(x) \{g^2(x)-1\} + f'(x) \cdot g(x) \{g^2(x)+1\}}{g^2(x)} dx$.

31. $\int_0^{\pi/4} \frac{\sin \theta + \cos \theta}{9 + 16 \sin 2\theta} d\theta$ 32. $\int_0^{\pi} \theta \sin^2 \theta \cos \theta d\theta$ 33. $\int_0^{\pi/2} \frac{1 + 2 \cos x}{(2 + \cos x)^2} dx$ 34. $\int_0^{\pi/2} \frac{x + \sin x}{1 + \cos x} dx$
35. Let $A = \int_{3/4}^{4/3} \frac{2x^2 + x + 1}{x^3 + x^2 + x + 1} dx$ then find the value of e^A .
36. $\int_0^1 \frac{2 - x^2}{(1+x)\sqrt{1-x^2}} dx$ 37. $\int_{-1}^1 \left(\frac{d}{dx} \left(\frac{1}{1+e^{1/x}} \right) \right) dx$ 38. $\int_1^e \frac{dx}{\ln(x^x e^x)}$
39. $\int_0^{\pi} \left[\cos^2 \left(\frac{3\pi}{8} - \frac{x}{4} \right) - \cos^2 \left(\frac{11\pi}{8} + \frac{x}{4} \right) \right] dx$ 40. If $f(\pi) = 2$ & $\int_0^{\pi} (f(x) + f''(x)) \sin x dx = 5$, then find $f(0)$
41. $\int_a^b \frac{|x|}{x} dx$ 42. $\int_{\ln 2}^{\ln 3} f(x) dx$, where $f(x) = e^{-x} + 2e^{-2x} + 3e^{-3x} + \dots \infty$
43. $\int_0^{\pi/2} \sqrt{\frac{\sec x - \tan x}{\sec x + \tan x}} \cdot \frac{\operatorname{cosec} x}{\sqrt{1 + 2 \operatorname{cosec} x}} dx$ 44. $\int_0^1 x f''(x) dx$, where $f(x) = \cos(\tan^{-1} x)$
45. (a) If $g(x)$ is the inverse of $f(x)$ and $f(x)$ has domain $x \in [1, 5]$, where $f(1) = 2$ and $f(5) = 10$ then find the value of $\int_1^5 f(x) dx + \int_2^{10} g(y) dy$.
- (b) Suppose f is continuous, $f(0) = 0$, $f(1) = 1$, $f'(x) > 0$ and $\int_0^1 f(x) dx = \frac{1}{3}$. Find the value of the definite integral $\int_0^1 f^{-1}(y) dy$.

ANSWER KEY

1. $\frac{\pi^2}{4}$ 2. $\frac{1}{2} \ln \left(\frac{e}{2} \right)$ 3. $2\sqrt{e}$ 4. $2 - \frac{\pi}{2}(a - 3b)$ 5. $\frac{1}{6}$ 6. $e - \frac{2}{\ln 2}$ 7. $\frac{\pi}{4}$
8. $\ln \frac{4}{3}$ 9. $\frac{1}{6}$ 10. $\frac{\pi \ln 3}{2}$ 11. $\frac{\pi}{6}$ 12. $\frac{\sqrt{3}}{2} - 1 + \frac{\pi}{6}$ 13. $\frac{\pi - 3}{16}$
14. $\frac{2}{3} \tan^{-1} \frac{1}{3}$ 15. $\frac{\pi}{3}$ 16. $\frac{\theta}{\sin \theta}$ 17. $\frac{1}{2} \left(\frac{\pi}{6} + \ln 3 - \ln 2 \right)$ 18. $\frac{1}{3}$ 19. $\frac{3\pi}{2}$
20. $\frac{1}{2} \ln(2 + \sqrt{3})$ 21. $\frac{1}{4} \ln \frac{32}{17}$ 22. $\frac{1}{3} \frac{a^3 - b^3}{a^2 - b^2}$ 23. (a) $2(\sqrt{2} + 1)$; (b) $\left(\pi - \frac{\pi^2}{4} \right)$
24. $\frac{\pi}{4} \left(\frac{\pi}{4} - 1 \right) + \frac{1}{2} \ln 2$ 25. 13 26. $\frac{\alpha}{2 \sin \alpha}$ if $\alpha \neq 0$; $\frac{1}{2}$ if $\alpha = 0$ 27. 1
28. $\frac{1}{2} \ln 3$ 29. $\frac{3\pi + 8}{24}$ 30. 2009 31. $\frac{1}{20} \ln 3$ 32. $-\frac{4}{9}$ 33. $\frac{1}{2}$
34. $\frac{\pi}{2}$ 35. $\frac{16}{9}$ 36. $\frac{\pi}{2}$ 37. $\frac{2}{1+e}$ 38. $\ln 2$ 39. $\sqrt{2}$ 40. 3
41. $|b| - |a|$ 42. $\frac{1}{2}$ 43. $\pi/3$ 44. $1 - \frac{3}{2\sqrt{2}}$ 45. (a) 48; (b) $2/3$

Result JEE Main

Result Highlights 2019



100 Percentile
HIMANSHU GAURAV SINGH
2019 (*SDCCP)



99.99 Percentile
GAURAV KRISHAN GUPTA
2020 (DLP)



99.98 Percentile
SARTHAK ROUT
2020 (CCP)



99.98 Percentile
VIBHAV AGGARWAL
2019 (CCP)



99.97 Percentile
RITVIK GUPTA
2020 (DLP)



99.97 Percentile
BHAVYA JAIN
2020 (CCP)



99.96 Percentile
AYUSH PATTNAIK
2019 (CCP)



99.96 Percentile
SAYANTAN DHAR
2020 (DLP)

Result Highlights 2018



Sarthak Behera



Shashank Agrawal



Shreya Pathak



Raaghav Raaj



Aayush Kadam



Gunjan

Result Highlights 2017



Abhay Goyal



Piyush Tibarwal



Abhinav Kumar



Tushar Gautam



Saurav Sahoo



Pratyush

Mentored by Team



Kushal Babel
2014



Gourika Bang
2015



Amit Malav
2013



Suraj Sanjay Jog
2012



Chitraang Murdia
2014



Pulkit Tandon
2012



Kapil Shobhnani
2016



Utkarsh Sinha
2015



S John S
2015